

Chapter 1 Holomorphic Functions in Several Variables

1.1 Fundamentals

1.1.1 Cauchy-Riemann Equation

Definition 1.1.1.1 ...

1.1.2 Some of Conclusions

◇ **Lemma 1.1.2.1** ...

1.2 Theorems

1.2.1 Cauchy's Formula

♠ **Proposition 1.2.1.1** (Cauchy) If f is a function defined in a neighborhood U of a closed polydisc $\bar{\Delta}(a, r) = \prod D(a_n, r_n)$ and holomorphic in (contains) a neighborhood U and holomorphic in each variable z_i at each point U , then

$$f(z) = \left(\frac{1}{2\pi i}\right)^n \int_{|\xi_n - a_n|=r_n} \cdots \int_{|\xi_1 - a_1|=r_1} \frac{f(\xi_1, \dots, \xi_n)}{\prod (\xi_i - z_i)} d\xi_i, z \in \bar{\Delta}(a, r)$$

◦ **Proof** Omitted, from Cauchy's formula in one variable. □

Note that:

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♠ **Proposition 1.2.1.2** (Osgood's Lemma)

◦ **Proof** ... □

♠ **Proposition 1.2.1.3** (Cauchy's Inequality)

1.2.2 Hartog's Theorem

♠ **Proposition 1.2.2.1** (Jensen's Inequality)

◦ **Proof** ... □

♠ **Proposition 1.2.2.2** ...

◦ **Proof** ... □

1.2.3 Convergence Theorem

1.3 Domains of Holomorphy

Chapter 2 AG Course

Chapter 3 Coherent Analytic Sheaves

Chapter 4 Stein Space

Chapter 5 Frechet Sheaves, Cartan's Theorem

Chapter 6 Serre's GAGA Theorem